

NyumbaHub

Real Estate Marketplace — Kotlin Android App

Connecting property owners, renters & buyers

DOCUMENT TYPE

App Master Plan & Feature Blueprint

VERSION

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FOCUS

Residential: Sale, Rent & Subscription

NH

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Executive Summary & Vision

What we're building and why

What is NyumbaHub?

NyumbaHub ("Nyumba" = house in Swahili) is a Kotlin-native Android real estate marketplace that connects property owners, landlords, and real estate agents with buyers and renters through a clean, fast, and trusted platform. Phase 1 focuses entirely on **residential properties** — houses for sale and for rent — with a subscription model that allows listers to unlock premium exposure.

The Problem We Solve

- Fragmented listings — buyers waste time across WhatsApp groups, Facebook posts, and word-of-mouth
- No trust layer — no verification of properties or agents leads to fraud
- No standardised pricing data — renters and buyers negotiate blind
- Poor mobile experience — existing local platforms are web-first and not optimised for Android
- No structured search — location, price, bedroom count filters are missing or broken

Our Value Proposition

For Buyers/Renters	For Sellers/Landlords	For Agents
Browse verified listings	Post & manage listings	Subscription-based leads
Filter by location, price, beds	Reach thousands of seekers	Featured agent profiles
Direct in-app messaging	Subscription tiers for visibility	Analytics dashboard
Save & compare properties	Upload unlimited photos	CRM for inquiries
Schedule viewings	Mark as rented/sold instantly	Priority listing placement

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Market Research & Competitive Analysis

Learning from the world's best property platforms

Global Benchmark Apps

Before building, we studied five leading platforms across different markets to extract proven patterns, features that drive retention, and monetisation strategies that work at scale.

Platform	Country	Key Strength	Monetisation	Lesson for NyumbaHub
Zillow	USA	Zestimate AI valuation, rich map search	Agent ads + leads	Map-first search + price estimates
Rightmove	UK	Massive inventory, trusted brand	Agent subscriptions	Subscription model for listers
Property24	South Africa	African market context, mobile-first	Featured listings	Relevant local UX patterns
Bayut	UAE/MENA	Verified listings badge, TruCheck	Tiered agent plans	Verified badge builds trust
99.co	Singapore	Clean UI, commute-time filter	Premium listings	Lifestyle-fit filters

Key Patterns Extracted

- **Map-based search** is the #1 feature users expect — listings on a map with clustering
- **Verified listings** badge significantly increases inquiry rates (Bayut reports 3x uplift)
- **Photo quality** is the biggest driver of listing views — enforce minimum 5 photos
- **Subscription tiers** for listers (not commission) removes friction and scales revenue predictably
- **Saved searches + push alerts** bring users back daily — critical for retention
- **Price-per-sqm** transparency (Rightmove) builds buyer confidence
- **Agent profiles with reviews** establish trust and differentiate agents from private sellers
- **Quick-filter pills** (For Sale | For Rent | Price Range | Beds) above the list are universal UX

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App Architecture Overview

Android-first, scalable, clean architecture

Architecture Pattern — MVVM + Clean Architecture

NyumbaHub follows **MVVM (Model-View-ViewModel)** combined with **Clean Architecture layers**. This separation ensures the app is testable, maintainable, and scalable as features grow.

Layer	Responsibility	Key Components
Presentation	UI, user events, state observation	Jetpack Compose, ViewModels, UI State
Domain	Business logic, use cases	UseCases, Repository interfaces, Entities
Data	Data sources, mapping, caching	Repositories, Room DB, Retrofit, Firebase
Remote	API calls, cloud services	REST API, Firebase Auth, Google Maps SDK
Local	Offline cache, drafts	Room Database, DataStore Preferences

Key Architectural Decisions

- **Single Activity** with Jetpack Compose Navigation — no legacy Fragment stack
- **Repository Pattern** — data layer abstracted so we can swap Firebase for a custom API later
- **Offline-first** — Room caches listings so users browse without constant internet
- **Coroutines + Flow** — reactive, non-blocking data streams throughout the app
- **Hilt Dependency Injection** — clean, testable component wiring
- **Modular build** — :app, :feature-listings, :feature-search, :feature-chat, :core-ui, :core-network

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User Roles & Personas

Who uses the app and what they need

Four Core User Roles

■ Guest / Browser

- No account required for browsing
- Can view listings, photos, and location on map
- Cannot contact lister or save favourites
- Prompted to register when trying to interact
- Conversion goal: sign up within 3 sessions

■ Seeker (Buyer / Renter)

- Free account — email or Google sign-in
- Full search with filters and map view
- Save favourites and set up alerts
- Send inquiries and chat with listers
- Schedule and manage viewing appointments
- Review and rate agents after transactions

■ Lister (Private Seller / Landlord)

- Must register and verify phone number
- Post up to 3 listings on free tier
- Manage listing status: Available / Rented / Sold
- Receive and respond to inquiries via in-app chat
- Upgrade to paid subscription for more listings and features
- Access basic analytics: views, inquiries per listing

■ Agent / Agency (Premium)

- Verified agent badge after document check
- Subscription plan with unlimited listings
- Dedicated agency profile page with all listings
- Lead management dashboard (CRM-lite)

- Priority listing placement in search results
- Performance analytics: views, CTR, conversion rate

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Core Features — Property Listings

The heart of the platform

Listing Data Model

Every property listing captures the following structured data fields:

Field Group	Fields	Required?
Identity	listingId, createdAt, updatedAt, status (ACTIVE/RENTED/SOLD/PAUSED)	■ Yes
Type	listingType (SALE/RENT), propertyType (HOUSE/APARTMENT/TOWNHOUSE/VILLA)	■ Yes
Location	country, city, district, neighbourhood, latitude, longitude, mapPin	■ Yes
Pricing	price, currency, negotiable (bool), depositMonths, monthlyRent, paymentFreq	■ Yes
Property Details	bedrooms, bathrooms, toilets, sizeSqm, floors, yearBuilt, furnished	■ Yes
Features	parking, garden, security, borehole, solar, generator, pool, serviced	Optional
Media	photos[] (min 5, max 30), videoUrl, virtualTourUrl, floorPlanUrl	■ Min 5 photos
Description	title (max 80 chars), description (max 1000 chars), highlights[]	■ Yes
Contact	listerId, agentId, contactPhone, whatsapp, preferredContactMethod	■ Yes
Metadata	viewCount, inquiryCount, savedCount, featuredUntil, boostLevel	System auto

Listing Lifecycle & Statuses

- **DRAFT** — lister is filling in details, not visible to public yet
- **PENDING_REVIEW** — submitted, awaiting automated + manual checks
- **ACTIVE** — visible in search, receiving inquiries
- **FEATURED** — active + boosted placement from subscription or one-time boost
- **PAUSED** — lister temporarily hides listing (e.g. on holiday)
- **RENTED / SOLD** — transaction complete, listing archived but viewable
- **EXPIRED** — free listings expire after 30 days unless renewed
- **REJECTED** — failed moderation with reason provided to lister

Photo & Media Requirements

- Minimum 5 photos required before a listing can go ACTIVE
- Maximum 30 photos per listing

- Auto-compress on upload: max 1200px wide, quality 85% JPEG
- First photo auto-selected as cover thumbnail (user can reorder)
- Optional: YouTube/video link for walkthrough
- Optional: Floor plan image upload
- AI-powered duplicate photo detection flags repeated uploads

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Core Features — Search & Discovery

How users find properties

Search Modes

- **Text Search** — keyword search on title, neighbourhood, city
- **Map Search** — Google Maps view with listing pins, cluster at zoom-out
- **Nearby Search** — GPS-based "Properties near me" with radius slider
- **Saved Search** — user names a filter set, receives push notifications on new matches

Filter Parameters

Filter	Type	Options / Range
Listing Type	Toggle	For Sale For Rent
Property Type	Multi-select chips	House, Apartment, Townhouse, Villa
Price	Range slider + manual input	Currency-aware min/max
Bedrooms	Stepper / chips	Studio, 1, 2, 3, 4, 5+
Bathrooms	Stepper	1, 2, 3, 4+
Size (sqm)	Range slider	Min/max square metres
Location	City > District > Neighbourhood	Hierarchical picker
Amenities	Checkbox grid	Parking, Garden, Pool, Solar, Security, etc.
Furnished	Toggle	Furnished Unfurnished Semi
Listed by	Toggle	Owner Agent Any
Posted within	Dropdown	Today, Last 7 days, Last 30 days
Verified only	Toggle switch	Show TrustMark-verified listings only

Sort Options

- Price: Low to High
- Price: High to Low
- Newest First
- Most Popular (views)
- Nearest (distance from current location)
- Featured First

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Core Features — Messaging & Inquiries

Safe communication between seekers and listers

Inquiry System

All communication happens inside the app — phone numbers are hidden until the lister chooses to reveal them. This protects privacy and keeps the interaction traceable for safety.

Inquiry Flow

- 1. Seeker taps 'Send Inquiry' on a listing
- 2. Pre-filled form: name, phone, message template + custom message
- 3. Lister receives push notification
- 4. Both parties enter an in-app chat thread tied to that listing
- 5. Lister can reveal phone number / WhatsApp voluntarily from chat
- 6. Viewing appointment can be scheduled from within the chat
- 7. After visit, seeker is prompted to leave a rating for the agent/lister

Chat Features

- **Text messaging** — real-time via Firebase Realtime DB or XMPP
- **Photo sharing** — seeker can share photos of damage / lister can share more property images
- **Quick replies** — lister can set auto-reply templates ("Thanks for your interest...")
- **Read receipts** — single tick (sent) / double tick (read)
- **Viewing scheduler** — calendar picker inside chat, generates confirmation for both parties
- **Report & Block** — any user can report abusive messages; automatic flagging after 3 reports
- **Archived chats** — completed inquiries auto-archive after listing is marked RENTED/SOLD

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Core Features — Subscriptions & Monetisation

Revenue model inspired by Rightmove & Bayut

Subscription Tiers for Listers

Feature	Free	Basic	Pro	Agent
Monthly Price	FREE	\$5	\$15	\$40
Active Listings	3	10	30	Unlimited
Photos per Listing	10	20	30	30
Featured Placement	■	■	■ 2/month	■ 5/month
Analytics Dashboard	Basic	Standard	Advanced	Full + Export
Verified Badge	■	■	■	■
Priority in Search	■	■	■	■ Top Priority
WhatsApp Integration	■	■	■	■
Lead Notifications	Email only	Push + Email	Push + Email + SMS	Push + Email + SMS
Listing Duration	30 days	60 days	90 days	Continuous
Agent Profile Page	■	■	■	■

Additional Revenue Streams

- **Listing Boost** — one-time fee to push a single listing to the top for 7 days (\$2)
- **Premium Photography** — partner with local photographers, app earns referral fee
- **Home Valuation Report** — AI-assisted price estimate report, sold as one-time purchase (\$3)
- **Moving Services** — affiliate links to removal companies when listing is marked SOLD/RENTED
- **Banner advertising** — Phase 2+ only, never on free-tier user feeds to avoid annoyance

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Kotlin Tech Stack & Project Structure

Every library and tool we'll use

Android / Kotlin Stack

Category	Technology	Purpose
Language	Kotlin 1.9+	Primary language, 100% Kotlin
UI Framework	Jetpack Compose	Declarative UI, Material Design 3
Navigation	Compose Navigation	Type-safe navigation with deep links
DI	Hilt (Dagger)	Dependency injection throughout app
Async	Kotlin Coroutines + Flow	Non-blocking operations, reactive streams
Networking	Retrofit 2 + OkHttp	REST API calls, interceptors for auth
JSON	Gson / Moshi	Serialise/deserialise API responses
Local DB	Room Database	Offline cache for listings and messages
Preferences	DataStore	User settings, tokens, filter prefs
Images	Coil 3	Async image loading, caching, thumbnail
Maps	Google Maps SDK + Places	Map search, geocoding, place picker
Auth	Firebase Authentication	Email, Google, Phone OTP sign-in
Push Notifications	Firebase Cloud Messaging (FCM)	New inquiries, price drops, alerts
Real-time Chat	Firebase Realtime Database	In-app messaging threads
File Storage	Firebase Storage	Photo and document uploads
Backend API	Firebase Firestore (Phase 1)	Listings, users, subscriptions data
Payments	Flutterwave / Pystack SDK	Mobile money + card payments
Analytics	Firebase Analytics + Crashlytics	Usage tracking, crash reporting
Testing	JUnit5, MockK, Turbine	Unit tests, Flow testing
CI/CD	GitHub Actions + Firebase App Distribution	Automated builds and QA distribution

Module Structure

- :app — main entry point, DI graph, navigation host
- :core:ui — shared Compose components, theme, typography, colours
- :core:network — Retrofit client, auth interceptor, response models
- :core:data — repositories, Room DB, DataStore, Firebase services

- :core:domain — use cases, entity models, repository interfaces
- :feature:auth — login, register, OTP verification, profile setup
- :feature:listings — browse, listing detail, photo gallery
- :feature:search — search bar, filters, map view, saved searches
- :feature:post — create/edit listing form, photo upload, preview
- :feature:chat — inquiry list, chat thread, viewing scheduler
- :feature:subscription — plans screen, payment flow, manage subscription
- :feature:profile — user/agent profile, my listings, favourites, settings

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UI/UX Design System

Visual language, screens & user flows

Design Principles

- **Mobile-first** — designed for 360–414dp screen widths; tested on small and large phones
- **Speed** — listings load skeleton screens in <100ms; photos lazy-load progressively
- **Clarity** — price, beds, baths, and location visible on every card without tapping
- **Trust signals** — verified badge, listing age, lister response rate all on detail screen
- **One-thumb navigation** — primary actions within thumb reach; bottom navigation bar

Core Screen Inventory

Screen	Primary Purpose	Key UI Elements
Splash / Onboarding	Brand intro + value prop (3 slides)	Lottie animation, skip button
Home Feed	Personalised listing cards	Quick-filter pills, category chips, featured banner
Search Results	Filtered listing grid or list	Sort bar, filter button, map toggle, count badge
Map View	Listings on Google Map	Clustered pins, price bubbles, bottom sheet preview
Listing Detail	Full property info	Photo carousel, key stats, description, CTA buttons
Photo Gallery	Full-screen photos	Swipe gallery, pinch-to-zoom, counter
Inquiry / Chat	Communicate with lister	Chat thread, quick-reply, viewing scheduler
Post Listing	Create a new property listing	Multi-step form, photo upload, map location picker
My Listings	Manage posted properties	Status chips, edit/pause/delete, analytics preview
Subscription Plans	Upgrade account tier	Tier comparison cards, payment button
Profile & Settings	Account management	Avatar, stats, saved searches, log out
Notifications	Alerts and messages	Categorised: Inquiries, Price drops, New matches
Favourites	Saved listings	Listing cards, compare toggle, delete swipe
Agent Profile	Agent/agency public page	Bio, listing count, reviews, all active listings

Colour Palette & Typography

- Primary: Deep Navy #1A3C5E — headers, toolbar, primary buttons
- Accent: Vibrant Orange #E87722 — CTAs, badges, highlights, icons
- Success: Forest Green #2E7D32 — ACTIVE status, verified badge
- Error: Deep Red #C62828 — expired, rejected status indicators
- Background: #F4F7FB (light), #FFFFFF (card surfaces)
- Typography: Poppins (headers, weights 600/700), Roboto (body, 400/500)
- Icon set: Material Symbols Rounded — consistent 24dp icons throughout

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Backend & APIs

Server-side services and external integrations

Phase 1 Backend: Firebase-First

Phase 1 uses Firebase as the primary backend to accelerate development and reduce DevOps overhead. The architecture is designed so that Firebase can be swapped for a custom Node.js/Kotlin Spring Boot backend in Phase 3 without refactoring the Android data layer (repository pattern ensures this).

Firestore Services Used (Phase 1)

- Authentication — email, Google sign-in, phone OTP for listers
- Firestore — listings, users, agents, subscriptions, inquiries documents
- Realtime Database — live chat messages (lower latency than Firestore for messaging)
- Storage — property photos, floor plans, agent profile photos
- Cloud Functions — listing expiry cron, subscription renewal webhooks, moderation triggers
- FCM — push notifications for new inquiries, matches, and reminders
- App Check — prevents API abuse from non-authentic app clients

External API Integrations

Integration	Provider	Usage
Maps & Geocoding	Google Maps Platform	Map view, place search, reverse geocoding for address
Payments	Flutterwave	Card payments, Mobile Money (MTN, Airtel), bank transfer
SMS OTP	Africa's Talking / Twilio	Phone number verification for listers
Image Moderation	Google Vision AI / AWS Rekognition	Auto-detect inappropriate images on upload
Email	SendGrid	Transactional emails: welcome, inquiry receipt, subscription
Analytics	Firebase Analytics	User funnels, screen time, feature usage
Crash Reporting	Firebase Crashlytics	Real-time crash alerts with stack traces

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Database Schema

Core data structures (Firestore collections)

Firestore Collections Overview

Collection	Document ID	Key Fields
users	userId (Firebase UID)	displayName, email, phone, role, avatarUrl, createdAt, isVerified
listings	listingId (UUID)	title, type, price, location{}, specs{}, media[], status, listerId, viewCount, featuredUntil
agents	agentId (userId)	agencyName, licenseNo, verifiedAt, bio, reviewScore, listingCount
inquiries	inquiryId	listingId, seekerId, listerId, status (OPEN/REPLIED/CLOSED), createdAt
messages	messageId	inquiryId, senderId, text, mediaUrl, sentAt, readAt
subscriptions	userId	plan (FREE/BASIC/PRO/AGENT), startDate, endDate, paymentRef, autoRenew
payments	paymentId	userId, amount, currency, method, status, reference, createdAt
savedListings	userId+listingId	userId, listingId, savedAt
savedSearches	searchId	userId, filters{}, name, alertEnabled, lastAlertAt
reviews	reviewId	agentId, reviewerId, rating (1-5), comment, listingId, createdAt
notifications	notifId	userId, type, title, body, data{}, readAt, createdAt
reports	reportId	reporterId, targetType, targetId, reason, status (PENDING/RESOLVED)

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Security & Privacy

Protecting users and their data

Security Measures

- **Firebase App Check** — only genuine app installs can hit the backend
- **Firestore Security Rules** — users can only read/write their own documents; listers cannot see other listers' analytics
- **Phone Verification Required** — mandatory for posting listings (reduces fake listings)
- **Image Moderation** — AI scans every uploaded photo for inappropriate content before it goes live
- **Rate Limiting** — Cloud Functions cap inquiry sends to 10/hour per user to prevent spam
- **JWT token rotation** — Firebase ID tokens expire every hour, refreshed silently
- **API Key restrictions** — Google Maps and other keys restricted to app package name and certificate
- **SSL Pinning** — OkHttp certificate pinning for production builds to prevent MITM attacks
- **Obfuscation** — R8/ProGuard full mode for release builds

Privacy Policy Essentials

- Phone numbers hidden from seekers until lister explicitly shares
- Users can delete their account and all associated data (GDPR-compliant flow)
- Location used only for search proximity — never stored without consent
- Analytics are aggregate only — no individual tracking sold to third parties
- Clear cookie/data use disclosure on first launch
- Compliance checklist: GDPR, Kenya Data Protection Act 2019 (as baseline)

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Development Roadmap

4 phases from MVP to full platform

PHASE 1

Foundation & MVP

Months 1–3

- ✓ Project setup: Gradle multi-module, Hilt, Compose, Room, Retrofit
- ✓ Firebase project setup: Auth, Firestore, Storage, FCM
- ✓ User auth flow: Register, Login (email + Google), OTP phone verification
- ✓ Core data models: User, Listing, Location, Agent entities
- ✓ Post listing flow: Multi-step form, photo upload (min 5), location picker
- ✓ Listing detail screen: Photo carousel, specs, description, map pin
- ✓ Basic home feed: Listing cards, category filter pills (Sale/Rent)
- ✓ Basic search: Text search + city/district filter + price range
- ✓ Inquiry system: Send inquiry form + simple in-app chat (Firestore RTDB)
- ✓ Profile screen: My listings, edit profile, account settings
- ✓ Internal testing: Firebase App Distribution to QA team

PHASE 2

Search Power & Subscriptions

Months 4–6

- ✓ Google Maps integration: Map view with listing pins, cluster, price bubbles
- ✓ Advanced filters: Beds, baths, size, amenities, furnished, agent/owner
- ✓ Saved searches + push notification alerts on new matches
- ✓ Full subscription system: Plan comparison, Flutterwave payment, tier gating
- ✓ Featured listings: Boost placement logic in search algorithm
- ✓ Listing analytics: Views, inquiries, save count per listing for lister
- ✓ Agent profile pages: Bio, reviews, all active listings
- ✓ Review & rating system: Post-viewing prompt, aggregate score display
- ✓ Moderation system: Image AI check, report button, admin review queue
- ✓ Photo gallery: Full-screen swipe, pinch-to-zoom, share button
- ✓ Play Store Alpha release to limited group

PHASE 3

Trust & Polish

Months 7–9

- ✓ TrustMark verification: Document upload, manual review, verified badge
- ✓ Viewing appointment scheduler inside chat thread
- ✓ Price drop alerts: Notify saved-listing users on price reduction
- ✓ Favourites comparison: Compare up to 3 properties side-by-side

✓ Lister quick-reply templates: Set auto-response messages
✓ Deep linking: Share listing URL that opens app to correct screen
✓ Listing renewal reminders: Push + email before 30-day expiry
✓ Performance optimisation: Paging 3 library, image caching audit
✓ Accessibility audit: TalkBack support, contrast ratios, touch target sizes
✓ Play Store Beta release (open testing track)

PHASE 4	Scale & Full Launch	Months 10–12
✓ Production backend migration: Custom Node.js/Kotlin Spring API (optional)		
✓ Home valuation estimator: AI price estimate based on location + specs		
✓ Moving services affiliate integration		
✓ Full analytics dashboard: Conversion funnels, retention, revenue charts		
✓ Agent CRM dashboard: Lead pipeline, notes on inquiries, follow-up reminders		
✓ Multi-language support: English + Swahili (i18n framework)		
✓ Tablet/large-screen layout: Adaptive Compose layouts for tablets		
✓ Play Store Production full release		
✓ App Store (iOS) feasibility assessment — Kotlin Multiplatform evaluation		

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Feature Checklist — Build Tracker

Use this to audit what's built vs. missing

Print or maintain this checklist digitally. For each item check the STATUS column as you build. Use it to compare your current build against this master plan.

■ Authentication

#	Feature	Status	Notes
1	Email/password registration and login	■ Todo ■ In Progress ■ Done	
2	Google Sign-In integration	■ Todo ■ In Progress ■ Done	
3	Phone number OTP verification for listers	■ Todo ■ In Progress ■ Done	
4	Password reset flow	■ Todo ■ In Progress ■ Done	
5	Profile completion after first sign-in	■ Todo ■ In Progress ■ Done	
6	Session management with token refresh	■ Todo ■ In Progress ■ Done	

■ Listing Management

#	Feature	Status	Notes
1	Multi-step listing creation form	■ Todo ■ In Progress ■ Done	
2	Photo upload (min 5, max 30, auto-compress)	■ Todo ■ In Progress ■ Done	
3	Location picker with Google Maps pin	■ Todo ■ In Progress ■ Done	
4	Price and listing type (Sale/Rent) selection	■ Todo ■ In Progress ■ Done	
5	Bedroom/bathroom/size input	■ Todo ■ In Progress ■ Done	
6	Amenities checklist	■ Todo ■ In Progress ■ Done	
7	Description and title fields	■ Todo ■ In Progress ■ Done	
8	Preview before submit	■ Todo ■ In Progress ■ Done	
9	Listing status management (Active/Paused/Sold/Rented)	■ Todo ■ In Progress ■ Done	
10	Edit existing listing	■ Todo ■ In Progress ■ Done	
11	Delete listing with confirmation	■ Todo ■ In Progress ■ Done	
12	Listing expiry and renewal prompt	■ Todo ■ In Progress ■ Done	

■ Search & Discovery

#	Feature	Status	Notes
1	Text search on title and location	■ Todo ■ In Progress ■ Done	
2	Filter: listing type (Sale/Rent)	■ Todo ■ In Progress ■ Done	
3	Filter: property type (House/Apartment/etc.)	■ Todo ■ In Progress ■ Done	
4	Filter: price range slider	■ Todo ■ In Progress ■ Done	
5	Filter: bedrooms and bathrooms	■ Todo ■ In Progress ■ Done	
6	Filter: amenities checkboxes	■ Todo ■ In Progress ■ Done	

7	Filter: furnished toggle	■ Todo ■ In Progress ■ Done	
8	Filter: listed by (owner/agent)	■ Todo ■ In Progress ■ Done	
9	Sort: price, date, popularity, distance	■ Todo ■ In Progress ■ Done	
10	Map view with listing pins and clustering	■ Todo ■ In Progress ■ Done	
11	Nearby search using GPS	■ Todo ■ In Progress ■ Done	
12	Saved searches with push alerts	■ Todo ■ In Progress ■ Done	

■ Listing Detail & Media

#	Feature	Status	Notes
1	Full-screen photo carousel	■ Todo ■ In Progress ■ Done	
2	Pinch-to-zoom on photos	■ Todo ■ In Progress ■ Done	
3	Property specs summary row (beds/baths/size)	■ Todo ■ In Progress ■ Done	
4	Description with expandable overflow	■ Todo ■ In Progress ■ Done	
5	Map pin showing property location	■ Todo ■ In Progress ■ Done	
6	Lister info card (name, response rate, rating)	■ Todo ■ In Progress ■ Done	
7	Verified badge display	■ Todo ■ In Progress ■ Done	
8	Share listing button (deep link)	■ Todo ■ In Progress ■ Done	
9	Save to favourites heart icon	■ Todo ■ In Progress ■ Done	
10	Send inquiry / Call CTA buttons	■ Todo ■ In Progress ■ Done	

■ Messaging & Inquiries

#	Feature	Status	Notes
1	Send inquiry from listing detail	■ Todo ■ In Progress ■ Done	
2	In-app chat thread per inquiry	■ Todo ■ In Progress ■ Done	
3	Real-time message delivery	■ Todo ■ In Progress ■ Done	
4	Read receipts (sent/read tick)	■ Todo ■ In Progress ■ Done	
5	Photo sharing in chat	■ Todo ■ In Progress ■ Done	
6	Quick reply templates for listers	■ Todo ■ In Progress ■ Done	
7	Viewing appointment scheduler in chat	■ Todo ■ In Progress ■ Done	
8	Report message / block user	■ Todo ■ In Progress ■ Done	
9	Push notification for new messages	■ Todo ■ In Progress ■ Done	

■ Subscriptions & Payments

#	Feature	Status	Notes
1	Subscription plans screen (Free/Basic/Pro/Agent)	■ Todo ■ In Progress ■ Done	
2	Feature comparison per plan	■ Todo ■ In Progress ■ Done	
3	Flutterwave payment integration	■ Todo ■ In Progress ■ Done	
4	Mobile money payment support	■ Todo ■ In Progress ■ Done	

5	Payment confirmation screen	■ Todo ■ In Progress ■ Done	
6	Subscription status in profile	■ Todo ■ In Progress ■ Done	
7	Listing boost one-time purchase	■ Todo ■ In Progress ■ Done	
8	Featured placement logic in search	■ Todo ■ In Progress ■ Done	

■ Profiles & Accounts

#	Feature	Status	Notes
1	User profile screen (avatar, name, stats)	■ Todo ■ In Progress ■ Done	
2	My Listings tab with status filters	■ Todo ■ In Progress ■ Done	
3	Favourites tab with saved properties	■ Todo ■ In Progress ■ Done	
4	Saved searches tab	■ Todo ■ In Progress ■ Done	
5	Notification preferences	■ Todo ■ In Progress ■ Done	
6	Account deletion flow	■ Todo ■ In Progress ■ Done	
7	Agent profile page (public view)	■ Todo ■ In Progress ■ Done	
8	Review and rating submission	■ Todo ■ In Progress ■ Done	
9	Agent reviews list view	■ Todo ■ In Progress ■ Done	

■ Notifications

#	Feature	Status	Notes
1	Push notification: new inquiry received	■ Todo ■ In Progress ■ Done	
2	Push notification: new message in chat	■ Todo ■ In Progress ■ Done	
3	Push notification: saved search match	■ Todo ■ In Progress ■ Done	
4	Push notification: price drop on saved listing	■ Todo ■ In Progress ■ Done	
5	Push notification: listing expiry reminder	■ Todo ■ In Progress ■ Done	
6	Push notification: subscription renewal	■ Todo ■ In Progress ■ Done	
7	In-app notification centre screen	■ Todo ■ In Progress ■ Done	

■■ Trust & Safety

#	Feature	Status	Notes
1	Image AI moderation on upload	■ Todo ■ In Progress ■ Done	
2	Report listing button	■ Todo ■ In Progress ■ Done	
3	Report user/message button	■ Todo ■ In Progress ■ Done	
4	Verified lister badge (document check flow)	■ Todo ■ In Progress ■ Done	
5	Firebase App Check	■ Todo ■ In Progress ■ Done	
6	Firestore security rules (per-user access)	■ Todo ■ In Progress ■ Done	
7	Rate limiting on inquiry sends	■ Todo ■ In Progress ■ Done	
8	Admin moderation queue (internal tool)	■ Todo ■ In Progress ■ Done	

■■ Technical & Quality

#	Feature	Status	Notes
1	Multi-module Gradle project structure	■ Todo ■ In Progress ■ Done	
2	Hilt dependency injection	■ Todo ■ In Progress ■ Done	
3	MVVM + Clean Architecture layers	■ Todo ■ In Progress ■ Done	
4	Room database for offline cache	■ Todo ■ In Progress ■ Done	
5	Paging 3 for infinite scroll listings	■ Todo ■ In Progress ■ Done	
6	Compose Navigation with deep links	■ Todo ■ In Progress ■ Done	
7	Unit tests for all UseCases	■ Todo ■ In Progress ■ Done	
8	UI tests for critical user flows	■ Todo ■ In Progress ■ Done	
9	Firebase Analytics events tracked	■ Todo ■ In Progress ■ Done	
10	Firebase Crashlytics integrated	■ Todo ■ In Progress ■ Done	
11	R8/ProGuard obfuscation for release	■ Todo ■ In Progress ■ Done	
12	CI/CD pipeline (GitHub Actions)	■ Todo ■ In Progress ■ Done	
13	Play Store listing prepared (screenshots, description)	■ Todo ■ In Progress ■ Done	

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KPIs & Success Metrics

How we measure if the app is working

Metric	Target (Month 3)	Target (Month 6)	Target (Month 12)
Total Active Listings	500	3,000	15,000
Monthly Active Users (MAU)	2,000	15,000	60,000
Daily Active Users (DAU)	300	2,500	10,000
Listings-to-Inquiry Rate	10%	15%	20%
Avg. Photos per Listing	≥5	≥8	≥10
Paying Lister Conversion	2%	5%	10%
Monthly Revenue (MRR)	\$200	\$2,500	\$15,000
App Store Rating	≥4.0	≥4.2	≥4.5
Crash-free Sessions	≥97%	≥98.5%	≥99%
Chat Response Rate (listers)	40%	60%	75%
User Retention (D30)	15%	25%	35%

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Risks & Mitigations

What could go wrong and how we prevent it

Risk	Likelihood	Impact	Mitigation
Fake / fraudulent listings	High	High	Phone OTP mandatory, image AI, user reporting, manual review queue
Low initial listing supply (cold start)	High	High	Launch in one city first, recruit 50 agents pre-launch, offer 3 months free Pro
Firebase costs exceed budget at scale	Medium	High	Set Firebase budget alerts; plan custom backend migration at 50K MAU
Payment integration failure in target market	Medium	High	Integrate both Flutterwave and Paystack as fallbacks
Google Maps API cost spike	Medium	Medium	Cache geocoding results in Room; use Maps only when map screen is active
Low lister retention (free tier too limited)	Medium	Medium	Generous free tier (3 listings), user research after Month 1
Negative App Store reviews from fraud victims	Low	High	Fast moderation SLA (<24h), clear reporting flow, refund policy
Competitor launches similar app	Low	Medium	Focus on local trust (TrustMark verification) and superior UX

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Future Expansion

Phase 2+ features beyond the housing MVP

Phase 1 is houses ONLY. The architecture is designed to support future property and service categories. These are intentionally OUT OF SCOPE for Year 1 to maintain focus.

Category	Feature	Phase
Commercial	Offices, retail, warehouses, land for sale	Phase 2 (Year 2)
Short-stay	Holiday rentals, Airbnb-style nightly booking	Phase 2
New Developments	Off-plan property sales from developers	Phase 2
iOS App	Swift/Kotlin Multiplatform iOS version	Phase 3
Web Platform	Kotlin + React web listing management portal	Phase 3
Mortgage Tools	Affordability calculator, bank partner referrals	Phase 3
Moving Services	Full moving service booking integration	Phase 2
Smart Home Data	Energy ratings, internet speed data per area	Phase 3
AI Recommendations	ML-powered personalised listing recommendations	Phase 3
Multi-country	Expand to neighbouring countries with localisation	Phase 4

This master plan is a living document. Update it every quarter as you build, launch, and learn from your users. The checklist in Chapter 15 is your primary build-vs-plan comparison tool.

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